

Emmanuel Muhiga, Manufacturing Engineer

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PROFESSIONAL SUMMARY

Manufacturing Engineer with practical experience in CNC machining, robotics (Motoman, ABB) and automated PLC manufacturing systems. Proficient in various CAD tools such as AutoCAD, Creo, Inventor & SolidWorks, for 3D modelling, creating detailed engineering drawings, and generating BOMs. Proven ability to improve production efficiency, including implementing lean practices that reduced downtime by 27%.

Core Competencies:

Design & CAD: Autodesk Inventor | Creo Parametric | AutoCAD | SolidWorks | GD&T | FEA

Manufacturing & Automation: CNC Machining | Robotics (ABB, Yaskawa Motoman, FANUC) | PLC (Allen-Bradley) | Sheet metal fabrication | Microsoft Project | Dynamics AX

Programming: RobotStudio | RSLogix 5000 | Excel VBA | MasterCam | C++

WORK EXPERIENCE

Champion Moyer Diebel/Bi-Line Systems
Manufacturing Engineer

Jordan Station, Ontario, CA.
May 2024 – present

- Lead and support manufacturing improvement initiatives focused on increasing equipment reliability, implementing Lean practices that reduced downtime by 27%.
- Manage CapEx projects from concept to implementation, conducting ROI and payback analyses for new equipment, tooling, and automation designed to reduce labor-intensive processes.
- Develop and maintain process documentation, standard assembly work instructions, and visual management tools to support consistent operations and preventive maintenance.
- Troubleshoot and optimize manufacturing equipment, including CNC machines, Motoman robotic welding cells, and automated systems, to minimize operational downtime.
- Review, implement, and validate engineering changes (ECNs) using Creo and AutoCAD, ensuring equipment, and manufacturing processes are safely integrated into production.
- Design and implement tooling, jigs, and fixture improvements using Creo to enhance productivity, repeatability, and operator safety.
- Conduct root cause investigations using Quality tools (Pareto charts, PDCA, 5-Whys) to identify recurring equipment and process failures, implement corrective actions, and prevent recurrence.

- Optimized CNC machining operations by modifying G-code, optimizing cutting parameters, and performing tool and offset adjustments to maintain dimensional accuracy and surface finish.
- Troubleshoot and optimized PLCs (Allen Bradley) using RSLogix 5000, identifying and resolving logic errors, and communication within FMS cells.
- Conducted root-cause analyses and process improvement initiatives using Quality tools such as Ishikawa diagrams to enhance machine efficiency and reduce setup time.
- Assisted in automation upgrades and PLC-related troubleshooting in FMS cells including editing programs and sensor calibration.
- Developed preventive maintenance and 5S programs to extend equipment life and improve uptime.
- Developed and maintained process documentation, work instructions, and setup sheets to standardize operations and ensure consistent production quality.

EDUCATION

Bachelor of Technology (B.tech) - Manufacturing Engineering
McMaster University.

Advanced Diploma - Mechanical Engineering Technology
Niagara College.

ADDITIONAL INFORMATION

Status: Permanent Resident

References: Available Upon Request